SYNAPSE

Greenwashing detection in corporate ESG reports through NeuroSymbolic AI and Counterfactual Reasoning

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Abstract. Corporate greenwashing, the selective or misleading presentation of sustainability performance has become one of the critical semantic challenges in Environmental, Social, and Governance (ESG) reporting. Existing validation frameworks employ static rule-based approaches that cannot effectively address sophisticated greenwashing attempts. As a solution Large Language Models (LLMs) can be utilized to provide the necessary semantic reasoning capabilities, but they remain unreliable in critical claim verification due to hallucination. In this paper, we propose an integrated greenwashing detection framework called *Synapse* that employs Counterfactual Retrieval Augmented Generation (RAG) with NeuroSymbolic AI to address the shortcomings of the existing approaches. Synapse introduces Counterfactual RAG, a novel dual query retrieval mechanism. It retrieves both supporting and refuting information for the claims in a simultaneous manner. The symbolic validation layer using an OWL ontology with a HermiT reasoner and SWRL constraints provides a factual safety net against hallucinated verdicts. The ontology is converted to natural language prior to prompt injection enabling structured reasoning alongside LLM-based semantic analysis. Evaluated on 169 pairwise blind comparisons using Claude Sonnet 4 and GPT-4o as independent judges achieving 84% agreement and a Cohen's Kappa of 0.678 with Synapse preferred in 90 of 143 agreed, over baseline without RAG and ontology. Additionally, Synapse achieved an overall ROUGE-1 score of 0.382, showing strong grounding of evidence in verifying claims.

Keywords: ESG Reporting, Greenwashing Detection, NeuroSymbolic AI, Retrieval-Augmented Generation (RAG), Large Language Models, OWL Ontology

1 Introduction

Global industries are increasingly held accountable for their role in driving environmental crises such as climate change, greenhouse gas (GHG) emissions, biodiversity loss, and the overexploitation of natural resources. These challenges directly threaten food security, public health, and the long-term stability of ecosystems[1] Businesses

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are not only contributors to these risks but also face growing exposure to ecological risks through reputational damage, regulatory penalties, and reduced investor confidence when linked to environmentally harmful activities[2].

In response to these global challenges, governments and international organizations have introduced cooperative frameworks. The Paris Agreement (2015), adopted by 195 parties, set the long-term goal of achieving net-zero GHG emissions from human activity [3] That same year, 194 UN member states adopted the **Sustainable Development Goals (SDGs)**, a set of 17 objectives designed to address interlinked social, environmental, and economic issues[3]. Environmental, Social, and Governance (ESG) indicators have emerged as a critical tool to measure and disclose corporate sustainability performance unlike traditional financial metrics, ESG indicators evaluate environmental stewardship, social responsibility, and governance structures, thereby helping investors and stakeholders assess corporate resilience and alignment with the SDGs.

Current ESG reporting frameworks, however, often prioritize reputational disclosures over genuine sustainability, making them prone to greenwashing and highlighting the need for a unified reporting standard with adaptive intelligent based data validation. In order to portray a company as more environmentally friendly than it actually is, greenwashing involves emphasizing one positive behavior or a tiny portion of its operations while hiding all others that have detrimental effects on the environment [5] Greenwashing is fundamentally a semantic problem existing in the current ESG reporting paradigm.

A distributed ledger technology (DLT)-based system can address this gap to a certain level by providing a consensus, secure and immutable platform that guarantees all participants agree on the same state of the ledger, for data sharing [6] but does not inherently address greenwashing.

Furthermore, static rule-based validation techniques in current ESG reporting frameworks are unable to identify sophisticated greenwashing or handle the ever-changing intricacies of ESG. The majority of current systems use strict validation techniques, such as statistical anomaly detection, specified threshold comparisons, and token-based format verification ([7], [8]). As well, different organisations' use of diverse certification and reporting techniques has led to varying quality levels, with some results falling short of acceptable norms [6].

Recent studies have explored Large Language Models (LLMs) to categorize vague, misleading, rhetoric or fabricated claims but does not definitively detect greenwashing[9]. Despite LLM’s ability to address semantic nuances and cross category generalization, their application in greenwashing detection remains underdeveloped.

In addition, LLM hallucination has become a concerning research gap despite its versatility in addressing numerous semantic nuanced problems. According to [10], by conducting a causal mediation analysis and utilising logit lens, they have discovered knowledge enrichment and self-attention answer extraction can get compromised during inference and are the main components that contributes to LLM hallucination. Over the years, many solutions have been proposed and implemented, however there is a considerable absence of systematically retrieving or reasoning with refuting evidence which is essential for cross domain challenges such as greenwashing detection and other areas that requires factual, unbiased verification, contradiction analysis and ethical reasoning.

These limitations motivate three research questions: how an LLM-based system can accurately identify vague or selectively presented ESG claims (RQ1); how LLM hallucinations can be mitigated in greenwashing detection using NeuroSymbolic AI (RQ2); and how a RAG-based framework can effectively retrieve both supporting and refuting evidence simultaneously (RQ3).

Synapse addresses these three challenges through an integrated pipeline of three novel mechanisms. RQ1 is addressed by the application of LLMs as semantic reasoners. As the problem of vague and selectively disclosed ESG claims is essentially a semantic problem, where the language on the surface may look plausible but lack meaningful commitment, sole rule-based methods will not be adequate to address this problem. To resolve RQ2, a hybrid NeuroSymbolic validation layer uses an OWL ontology integrated with a HermiT reasoner to evaluate SWRL-encoded logical consistency constraints independently of the LLM's parametric knowledge. The ontology is converted into natural language prior to prompt injection to allow LLM to reason over structured factual grounding rather than relying solely on learned weights, effectively acting as a symbolic guardrail against hallucinated verdicts. To resolve RQ3, a novel Counterfactual RAG architecture generates dual independent queries upon receiving an ESG claim, one retrieving internal company sustainability reports (D+) and one retrieving external contradicting sources including NGO reports and news articles (D−), with strict metadata filtering and optimised 1000-token chunking to maintain source separation throughout retrieval.

The remainder of this paper is organised as follows: Section II Related Work, Section III describes the Synapse Approach, Section IV and Section V present results and discussion respectively, Section VI discusses challenges, limitations and future work. Finally, Section VII concludes the paper.

2 Related Work

2.1 DLT-Based ESG Reporting Frameworks

To improve the accuracy and trustworthiness of ESG reports, several of these frameworks use IoT-based infrastructures to collect environmental data in addition to utilising DLT([11], [7]). [11] introduces a consortium blockchain and IoT-based multi-layered ESG reporting framework with *corporate crowdsensing*. [8] uses Knowledge Discovery in Databases (KDD) to automate data analysis and extract valuable insights from datasets related to blockchain transactions and ESG metrics to increase the identification of anomalies. It employs rule based and pattern based data mining to detect outliers but lack semantic reasoning about *why* patterns exist or whether they represent *genuine sustainability or greenwashing*. [7] verifies the accuracy of ESG data using a *centralized blockchain gateway* with versioning smart contracts and DNAinspired fact-telling. However, this method cannot adapt to new ESG metrics without updating the rules manually.

Both [8] and [7] improve data integrity through anomaly detection or rule based comparison. Although both systems rely on a centralized authority for data analysis and validation which restricts scalability and adaptability in distributed systems.

The use of reputation or token-based rating systems to promote accurate and transparent ESG disclosures is another characteristic noted in the DLT based ESG reporting frameworks literature. [7] presents fact index tokens weighted by accuracy, consistency, and transparency, whereas [11] suggests a tokenised reputation score that stakeholders can inspect. As a result, these methods are not flexible enough for dynamic ESG environments, where it is crucial to continuously adapt to new standards, data sources, and sustainability indicators.

2.2 NeuroSymbolic Approaches

[12] cites *NeuroSymbolic AI* as a promising strategy to accelerate the transition to AI-powered sustainable solutions in supply chains. NeuroSymbolic AI can offer adaptive, explainable intelligence, which is an intelligent solution compared to the static validation rules in [7] 's fact-telling mechanism and [8]'s KDD pattern mining. [13] suggests a method for managing ESG metrics that is driven by ontologies. By encouraging ESG data reuse and sharing across organizational, domain, and technical boundaries using a unified, intelligible structure, this approach improves *semantic interoperability* of ESG data.

However, [12]'s advocacy for NeuroSymbolic AI in supply chain sustainability through theoretical sounding, lacks concrete implementation for detecting greenwashing. Similarly, [13]'s ontology driven approach addresses semantic interoperability through unified structures, does not discuss mechanisms for ontology information conversion into natural language.

2.3 Causal Interpretability and LLM Hallucination

State of the art Language Models (LMs) tend to generate seemingly confident similar to factual generation but are hallucinated through the parametric knowledge obtained from pre training [10]. Through understanding internal mechanisms of LMs [10] have discovered that mechanistic causes of hallucination are knowledge enrichment hallucination in MLP layer and answer extraction hallucination in upper layer attention heads. [10] proposed MHM method that amplifies relevant knowledge from MLP layers and penalizes attention head for giving wrong answers. They utilized *causal mediation analysis* and *logit lens* to determine malfunctioning components. [14] conducted a behavioral study of LLMs to identify its capability in generating causal arguments and counterfactual reasoning. Additionally, they constructed a causal DAG on the causal discovery. CPER ; a counterfactual path based explainable recommendation proposed by [15] utilizing two counterfactual reasoning methods; representation perspective and topological structures (reinforcement learning to learn path manipulation strategy). This method learns explainable path weights via counterfactual learning. Meanwhile,

[16] proposes SELF-RAG that enhances LLM output through retrieval and self-reflection by criticizing the output using *reflection-tokens* that confirms output's relevance. This approach mitigates *indiscriminate retrieval* in common RAG.

While these approaches have made significant strides in addressing LLM reliability, they exhibit a lack of a systematic mechanism for counterfactual thinking and resolve inconsistencies in real-time claim validation scenarios. Despite [16]'s sophisticated use of reflection tokens to mitigate indiscriminate retrieval, it operates primarily through self-assessment and does not perform external validation against authoritative knowledge or logical reasoning frameworks to enhance debating skill of LLM considering supporting and 'what-if' scenarios side by side.

2.4 LLM Based Greenwashing Detection

Most recently, research has started applying LLMs to the greenwashing detection task. [9] use cross-category generalization to detect vague, misleading, and fabricated ESG statements, proposing the A3CG dataset annotated with aspect-action pairs from 1,679 sustainability reports. Although this is an improvement, this method does not incorporate external facts into the model's reasoning process and thus falls short of conclusive greenwashing detection. This method is also based on supervised learning that fails to generalize to semantically diverse ESG topics.

3 Synapse Approach

3.1 System Overview

Synapse is a domain-aware greenwashing detection framework. It tackles the fundamental challenges of current ESG data validation methods such as the limitations of static rule based validation systems in dealing with semantically nuanced tasks and the unreliability of isolated LLMs because of hallucination in factual verification tasks. The framework combines three highly interlocked modules, Counterfactual RAG, a NeuroSymbolic validation layer, and a structured reasoning pipeline into a single end-to-end system.

A stakeholder can input the name of the company and the ESG topic using the web interface. The framework performs a simultaneous search for the supporting evidence from the company's sustainability reports and the refuting evidence from other sources like reports from NGOs and news articles. The two search tasks are distinguished at the metadata level to maintain the integrity of the sources. The searched information is then fed to the NeuroSymbolic validation layer. The information is then evaluated using an OWL ontology with SWRL rules and a HermiT reasoner for logical consistency. The knowledge base in the ontology is converted to natural language before the injection of the prompt. The ontology class heirachy used in the NeuroSymbolic validation layer is illustrated in Fig. 1.

This particular system addresses all three of these research questions: LLM semantic reasoning is designed to address ambiguous and selectively disclosed information

(RQ1), the symbolic ontology layer addresses the hallucination problem by providing logical constraints independent of the LLM (RQ2), and Counterfactual RAG solves the one-sided search problem of traditional RAG by generating disjoint supporting and refuting documents within one query cycle (RQ3).

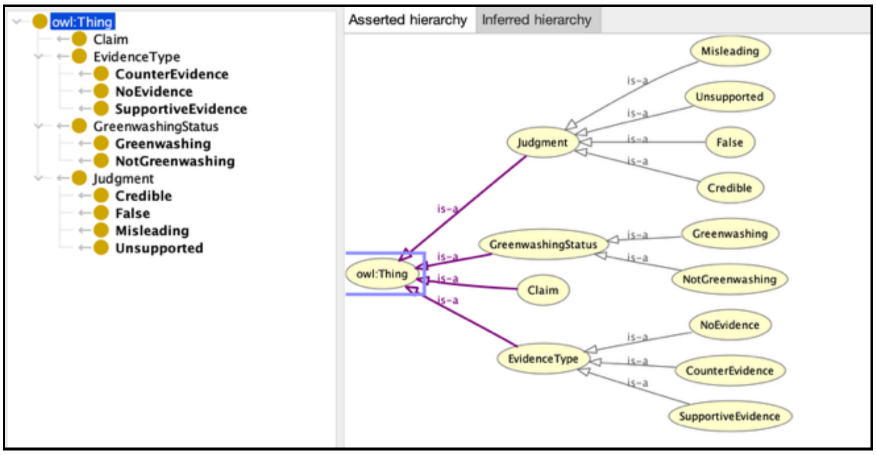

Fig. 1. Ontology Class Hierarchy in Protégé

3.2 System Architecture

The overall system architecture of Synapse is shown in Fig. 2. The three tier architecture design has been used to separate concerns related to presentation, logic and data. The Presentation Tier manages three UI components including Home Page UI for system navigation, Analyze Greenwashing UI for single company ESG claim analyzing, and Comparative Analytics Dashboard UI for multi company benchmarking. All components communicate with the backend via an API client.

The Logical Tier orchestrates greenwashing detection through three core modules coordinated by Fast API Gateway. The core modules include RAG Retrieval Logic; which generates supportive and counterfactual queries for LLM consumption, Symbolic Validation Logic; which loads OWL ontology converting ontology knowledge to natural language and Neural Validation Logic; which initializes Mistral 7B via Ollama ultimately invokes the LLM and parses structured JSON outputs containing classification and explanations.

The Data Tier offers persistent storage through two knowledge bases including FAISS Vector Store which performs text chunking and generation of the embeddings using HuggingFace's all-MiniLM-L6-v2, enables semantic similarity search, and persists to disk for Application Tier loading. The other knowledge base is the OWL Ontology uses to define the ESG classes, greenwashing patterns while enabling HermiT reasoner for consistency checking.

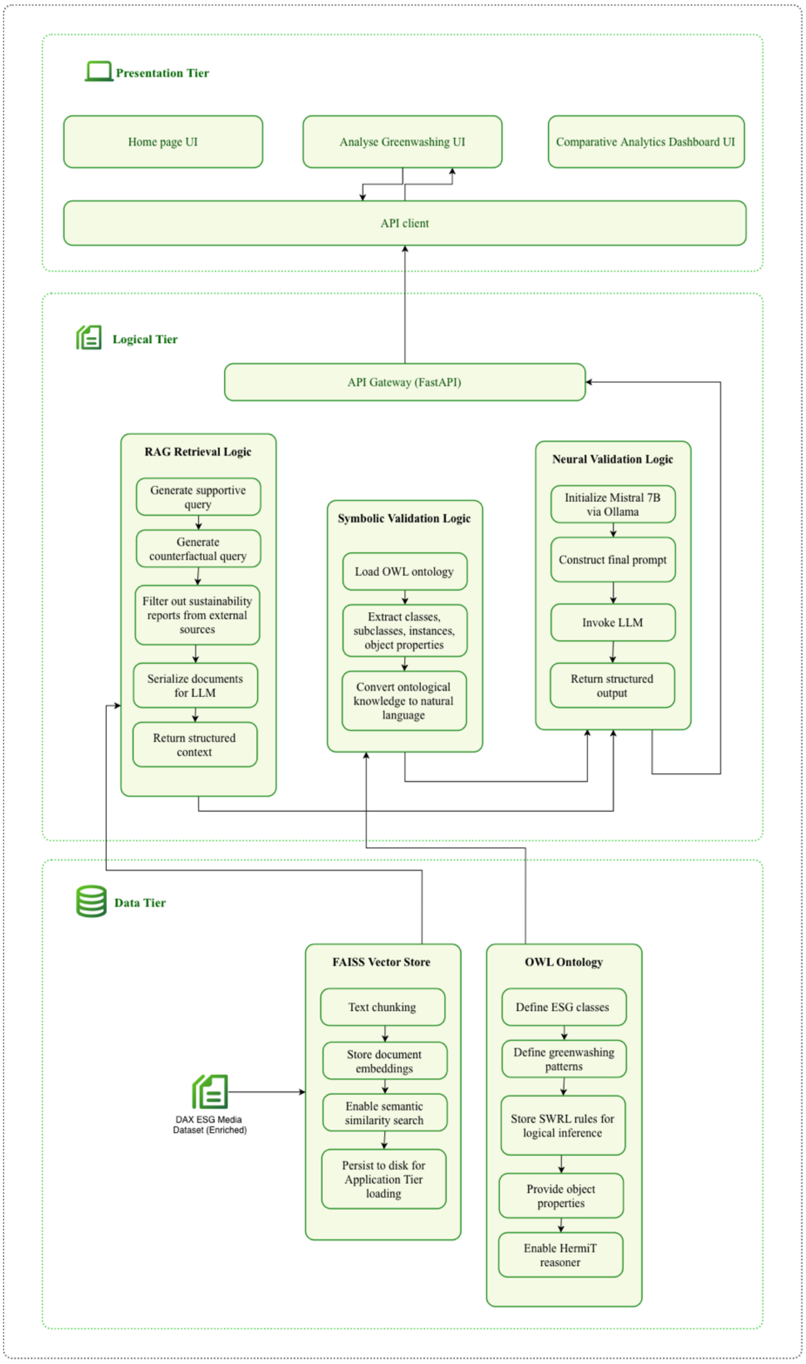

Fig. 2. Architecture Diagram

3.3 Sequence Diagram

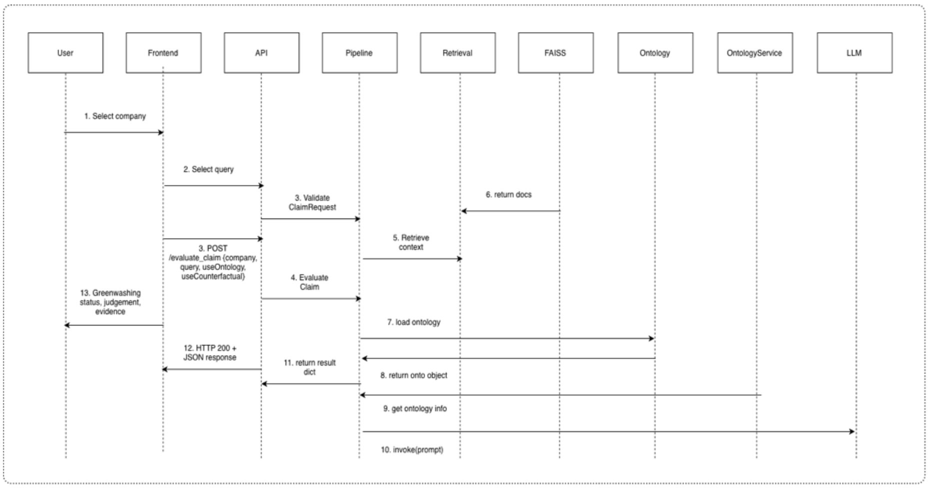

Fig. 3. Sequence Diagram

In Fig. 3. the sequence diagram illustrated in the time ordered sequence of messages passed between components in the process of greenwashing detection. The sequence diagram traces the entire flow from the user query to the final classification result. Additionally, it illustrates how the API Gateway manages the retrieval, reasoning using the ontology and validation by the LLM in the correct order.

3.4 Dataset and Preprocessing

The data preprocessing pipeline utilized in this study, helps extract sentences from corporate reports relevant to the ESG topics mentioned in the DAX ESG media dataset [17] using semantic similarity matching. DAX ESG Media Dataset consisting of about 11,000 documents in the English language that were about 38 German DAX companies in 2021-2023. It contains two types of documents; documents written by companies itself and the documents written by outside sources. There is a separate field with ESG topics that describes the document content such as GHGEmissions, FossilFuels and GenderEquality (Fig. 4.). There are total 40,136 ESG topics mentions and 359 unique topics with 72 most common topics (>100 mentions) and 103 rare topics (<10 mentions). According to [17], by integrating NLP capabilities with this dataset, will help identify ESG topics prone to greenwashing, and compare documents following standardized reporting frameworks (GRI, IFRS, SASB).

This data preprocessing pipeline was built using *zero-shot topic inferencing* workflow described in [18] and [19]. Finally, mapped ESG topic with ESG taxonomy (Fig. 5.) [20] into creating the enriched dataset with above extracted sentences/phrases.

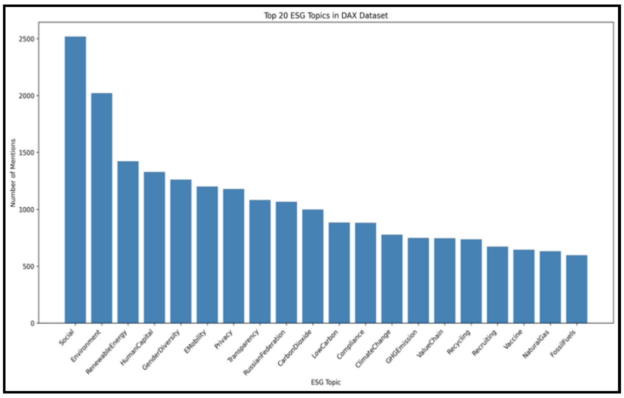

Fig. 4. DAX ESG Media Dataset most mentioned ESG Topics

```
taxonomy = {
    "Environmental": {
        "Air Emissions": ["GHG Emissions", "GHG Policies", "Non-GHG Air Emissions"],
        "Biodiversity": ["Biodiversity", "Forests"],
        "Environmental Management": ["Climate Risk Management", "Environmental Fines",
                                     "Environmental Management System", "Environmental Policy",
                                     "Environmental Reporting"],
        "Sustainable Production": ["Energy", "Hazardous Waste", "Packaging", "Toxic Spills",
                                   "Waste", "Water"],
        "Ozone-Depleting Gases": ["Ozone-Depleting Gases"],
        "GMOs": ["GMOs"],
        "Electromagnetic Fields": ["Electromagnetic Fields"],
        "Sustainable Product Development": ["Green Products", "Green Buildings", "Resource Efficiency"]
    },
    "Social": {
        "Human Rights": ["Child Labor", "Diversity", "Human Rights", "Indigenous Rights",
                         "Access to Basic Services", "Access to Healthcare", "Animal Welfare"],
        "Labour Practices": ["Employee Development", "Employee Turnover", "Health and Safety",
                             "Labor Practices", "Unions"],
        "Collective Bargaining": ["Collective Bargaining", "Remuneration"],
        "Community and Society": ["Community and Society", "Public Health", "Philanthropy"],
        "Customer Relationship": ["Customer Relationship"],
        "HIV Programs": ["HIV Programs"],
        "Clinical Trials": ["Clinical Trials"],
        "Product Safety": ["Product Safety"],
        "Responsible Marketing": ["Responsible Marketing"],
        "Supply Chain": ["Supply Chain"]
    },
    "Governance": {
        "Corporate Governance": ["Board", "Board Diversity", "Corporate Governance",
                                 "ESG Incentives", "Reporting Quality", "Sustainable Finance",
                                 "Chairperson-CEO Separation", "Audit"],
        "Business Ethics": ["Business Ethics", "Anti-competitive Practices", "Corruption",
                            "Privacy and IT"],
        "Global Compact": ["Global Compact Membership"],
        "Systemic Risk": ["Systemic Risk"],
        "Lobbying": ["Lobbying"],
        "Financial Inclusion": ["Financial Inclusion"],
        "Taxes": ["Taxes"],
        "Shareholders": ["Shareholders"],
        "Site Closure": ["Site Closure"]
    }
}
```

Fig. 5. ESG taxonomy [20]

During data preprocessing, after text cleaning and normalization, Semantic splitting was done to decompose documents into individual sentences for fine-grained semantic analysis. The text is split using NLTK's sentence tokenizer and produces manageable segments for embedding and topic matching. Then, the research employs WordNet to find semantically related terms. It restricts to first 2 synsets per word and top 5 synonyms total. Helps capture relevant sentences that use different terminology. Eg: Synonyms for "emissions" are "discharge", "release", "emanation". Then we identify the sentences that mention the ESG topic related terms from the document corpus by calculating term frequency-inverse document frequency (TF-IDF) for all n-grams (1-3 words). Sums TF-IDF scores across sentences and selects top N keywords. Finally, all the keywords are combined with topic words and synonyms. Following multi-component encoding, we converted all the components including topic name, topic + keywords, topic + context sentences, into dense semantic vectors using a pre-trained transformer *all-MiniLM-L6-v2 model*. Finally, we identify ESG-relevant sentences by computing semantic similarity between sentence and topic embeddings. Computes cosine similarity between each sentence vector and topic vector. Filters sentences above threshold. Finally, ranks by similarity score and selects top-K matches.

3.5 Counterfactual RAG and NeuroSymbolic Validation

The reasoning phase employs a novel Counterfactual RAG Inference algorithm (Algorithm 1) that systematically evaluates retrieved documents against a company's ESG claim.

Algorithm 1: Counterfactual RAG Inference

Require: Large Language Model *LLM,* Retrieval Evaluator *E,* Query Rewriter *W,* Prompt *P*
Input: Company name *C,* Query *Q, D* = $\{d_1, d_2, d_3, \ldots, d_k\}$ (Retrieved documents)
Output: Generated Response *R*
1 : $score_i = \begin{cases} +1, if\ E\ determines\ d_i\ supports\ the\ claim\ (C,Q) \\ -1, if\ E\ determines\ d_i\ refutes\ the\ claim\ (C,Q) \end{cases}$
2 : $D^+ = \{d_i \mid score_i = +1\}$
3 : $D^- = \{d_i \mid score_i = -1\}$
4 : $D^+_{topk} = TopK(D^+)$
5 : $D^-_{topk} = TopK(D^-)$
6 : $Internal\_Knowledge = k = Knowledge\_Refine(D^+_{topk}, D^-_{topk}, P, C, Q)$
7 : end
8 : $LLM\ Predicts\ R\ given\ k$ and C and Q

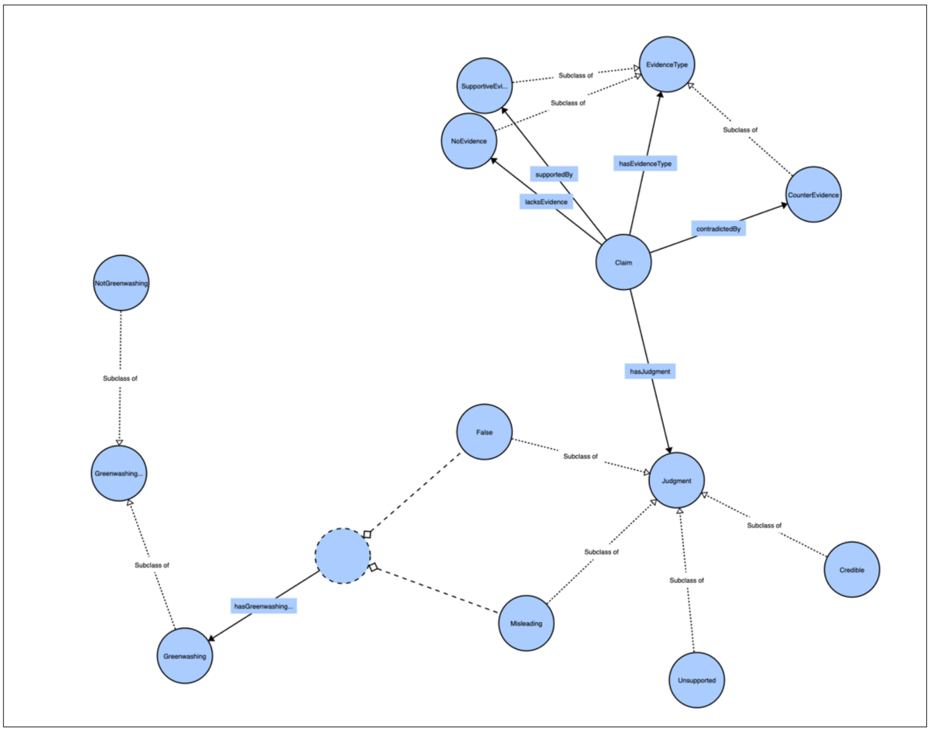

Fig. 6. Ontology WebVOWL visualization

Given the company name C and query (claim) Q, each retrieved document D is scored by a retrieval evaluator E as supporting ($score_i$= +1) or refuting ($score_i$= -1) while producing two disjoint document sets with topK selection D^{+}_{topk} and D^{-}_{topk} rather than defaulting to one sided retrieval as in conventional RAG[16]. Then this balanced context is passed to a NeuroSymbolic validation component. LLM, which plays as the neural subcomponent, conducts linguistic analysis of the claim against the retrieved evidence for spotting vague or misleading language patterns. The symbolic subcomponent utilizes an OWL ontology (Fig. 6.) with a HermiT reasoner [22] to enforce logical consistency constraints. It independently validates whether the claim is greenwashing or not. The outputs from both subcomponents are then aggregated into a structured JSON response with a greenwashing risk assessment.

The practical impact of this architecture is best illustrated through an end-to-end investor scenario, such as an inquiry into "Volkswagen" regarding "Air Emission".

Scenario 01: Volkswagen AG - Air Emissions
Input:


```
{
 "company": "Volkswagen AG",
 "query": "Air Emissions"
}
```

Output:

"result": {
"company_claim_summary": "The Volkswagen Group wants to measure the impact of its activities on air quality and plans to reduce production-related environmental externalities CO2, energy, water, waste, volatile organic compounds by 45 per vehicle compared to 2010.",
"object_property": "contradictedBy",
"judgment": "Misleading",
"greenwashing_status": "Greenwashing",
"reason_for_judgement": [
"Claim: 'The Volkswagen Group plans to reduce production-related environmental externalities CO2, energy, water, waste, volatile organic compounds by 45 per vehicle compared to 2010.' ([COMPANY_REPORT])",
"Counter-Evidence: 'The trial -- starting six years after the scandal broke -- is the first criminal case to target executives at VWs German headquarters who allegedly backed the idea of dodging emissions tests with a software trick.' ([EXTERNAL_SOURCE_COUNTERFACTUAL])",
"Despite the company's claim, there is an ongoing legal case involving emissions fraud, which casts doubt on their commitment to reducing air pollutants."
],
"summary_counter_evidence": "The Volkswagen diesel-fraud trial started over a scandal that won't go away ([EXTERNAL_SOURCE_COUNTERFACTUAL])"
}

4 Results

4.1 ROUGE Evaluation

We evaluated each system output using ROUGE metrics [21] to assess how faithfully the system grounds its generated outputs in retrieved evidence. ROUGE determines the quality of summary by measuring n-gram overlap between a generated text and a reference. The quality of the Synapse output is determined by concatenating the four key LLM output fields including company claim summary, summary counter evidence, summary support evidence, and reason for judgement. Higher ROUGE scores therefore indicate that the system output closely reflects retrieved evidence rather than hallucinated or paraphrased content.

ROUGE-1 (unigram overlap), ROUGE-2 (bigram overlap), ROUGE-L (longest common subsequence), and ROUGE-Lsum (sentence-level LCS) are the standard metrics that were reported as shown in the Table 1. System was tested and evluated on 241 ESG claims.

Table 1. Per sample and aggregate ROUGE scores for ontology guided greenwashing detection

Company	ESG Topic (Query)	GW Status	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum
Volkswagen AG	Air Emissions	Greenwashing	0.4128	0.2690	0.3488	0.3488
Bayer AG	Labour Practices	Greenwashing	0.6436	0.5455	0.6064	0.6064
Deutsche Telekom AG	Equality	Greenwashing	0.4935	0.4248	0.4481	0.4481
Deutsche Bank AG	Business Ethics	Greenwashing	0.4646	0.3932	0.4108	0.4108
Covestro AG	Corporate Governance	NotGreenwashing	0.2934	0.2023	0.2625	0.2625
Airbus SE	Business Ethics	NotGreenwashing	0.3026	0.1722	0.2566	0.2566
Puma SE	Air Emissions	Greenwashing	0.6020	0.4916	0.5151	0.5151
Porsche	Human Rights	Greenwashing	0.3904	0.2552	0.3425	0.3425
Zalando SE	Sustainable Production	NotGreenwashing	0.4045	0.2273	0.3146	0.3146
Adidas AG	Business Ethics	NotGreenwashing	0.3268	0.2980	0.3268	0.3268
Adidas AG	Corporate Governance	Greenwashing	0.3393	0.1796	0.1905	0.1905
Allianz SE	Sustainable Production	NotGreenwashing	0.2343	0.1941	0.2259	0.2259
BASF SE	Air Emissions	NotGreenwashing	0.4379	0.3114	0.3432	0.3432
Daimler AG	Sustainable Production	Greenwashing	0.3902	0.2857	0.3220	0.3220
⋮	⋮	⋮	⋮	⋮	⋮	⋮
Aggregate	All	—	0.3822	0.2619	0.3162	0.3165

4.2 LLM-as-Judge Evaluation

Using Claude Sonnet 4 and GPT-4o as independent LLM judges, authors evaluated 169 pairwise anonymous response comparisons between system A (RAG based with retrieval and ontology) and system B (baseline, without retrieval) eliminating label bias. The goal was to score each response on groundedness and consistency. LLM as judge results can be found in GPT vs Claude agreement confusion matrix in Fig. 7.

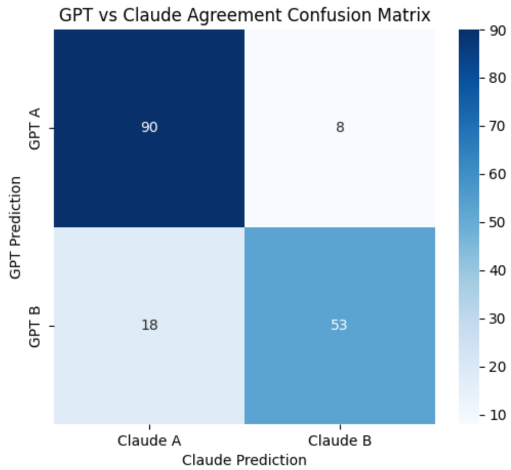

Fig. 7. GPT vs Claude Agreement Confusion Matrix

5 Discussion

According to the Table 1., The ROUGE-1 sum of 0.3822 and ROUGE-2 sum of 0.2619 show that the system's output corresponds to the evidence by copying words from it. The cases of greenwashing have higher ROUGE-1 scores of 0.47 compared to cases of non-greenwashing. This is because, with reliable and robust evidence available, the system copies words, for example Bayer AG with ROUGE-1 of 0.64 or Puma SE with ROUGE-1 of 0.60. However, when there is no robust evidence, the system must use only the company sustainability reports, which reduces the ROUGE-1 and ROUGE-2 scores. For example, Allianz SE has a ROUGE-1 of 0.23, and "Adidas AG Corporate Governance" has a ROUGE-L of 0.19. Therefore, it would be important to improve the accuracy of the system when the evidence is not robust.

According to the results of LLM-as-Judge in Fig. 7., two judges agreement on which system performed better has a rate of 84%. After LLM as judge evaluation we calculated Cohen's Kappa [23] using sklearn.metrics cohen_kappa_score to measure inter rater reliability and interpreted result using logical interpretation of Cohen's Kappa given in [23]. With a moderate agreement of 0.678, we can conclude that two judges making consistent judgements. At 90 cases, both judges preferred System A suggesting RAG and ontology based system produces more grounded assessments.

6 Challenges, Limitations and Future Work

Some of the challenges in the implementation were solved during the development process. The raw DAX ESG dataset lacked the mapping of Environment, Social, and Governance pillars. To solve this problem, the use of sentence-transformer-based semantic extraction and metadata normalization is applied in this research. FAISS retrieval was optimized from slow large chunk searches to 1000 token chunking with source-separated queries. The instability of LLM output causing JSON parsing errors was solved by applying fallback parsing logic.

Additionally, this research has a few limitations, leaving space for further improvements. The DAX ESG Media dataset has more company reports compared to external evidence that supports or refutes the company claim, which may create a biased judgement verdict due to evidence analysis, thereby affecting the accuracy of the judgement. Moreover, while the ROUGE score can efficiently measure the grounding of evidence, it may not fully capture the factuality of claim verification. Future studies should consider exploring other metrics such as FactScore, as well as a more balanced dataset with regards to company reports and other sources, in order to improve the accuracy of the judgement for greenwashing as well as non-greenwashing cases.

The current Synapse ontology includes foundational ESG class hierarchies and greenwashing pattern constraints that are currently sufficient for proof-of-concept validation, leaves an opportunity to build a comprehensive production-level ontology that is aligned to well-established regulatory reporting standards like GRI, SASB and IFRS. These could be incorporated directly into the OWL schema to enable the reasoner to detect not only semantic inconsistencies but also specific non-compliance to regulatory requirements. As a crucial step towards achieving *ESG dream*, a production-level greenwashing detection component could be integrated with an end-to-end DLT-based ESG reporting framework ensuring all key requirements including, security, transparency, and traceability of sustainability disclosures.

7 Conclusion

This paper proposes Synapse a framework for greenwashing detection in corporate ESG reports using NeuroSymbolic AI and Counterfactual reasoning. Further this paper introduces Counterfactual RAG, a fact checking mechanism, that retrieves both supporting and refuting information for company ESG claims. The experimental results demonstrate that Synapse produces more grounded and consistent outputs compared to baseline approaches without retrieval. A production level system can support investors and corporations make informed decisions about their sustainability performances while improving the transparency of ESG reports. However, as the dataset imbalance in the DAX ESG Media Dataset affects the accuracy of the system, using a balanced dataset is preferable. Future work will extend the ontology to align with regulatory standards such as GRI, SASB and IFRS.

Declaration on Generative AI

During the preparation of the manuscript, the authors used Grammarly for language refinement and grammar checking. The authors have critically reviewed and taken full responsibility for the publishing content. The ideas, arguments and results are entirely the work of human authors.

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